

Aleksander Kurgan

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Education

Stanford University

2026–2030 (Expected)

B.S. in Computer Science

- **Coursework (Expected 2026-27):** CS106B: Programming Abstractions (C++); MATH51: Linear Algebra & Multivariable Calculus; CS103: Math Foundations of Computing; CS107: Computer Organization & Systems.

Deep Run High School | GPA: 4.64 / 4.0 (weighted)

2022–2026

Center for Information Technology (CIT)

- **Coursework:** Dual Enrollment Discrete Mathematics, AP Computer Science A, AP Calculus BC, Enterprise Architecture, App Development; Computer Science Honor Society Council Member.

Work Experience

Academic Researcher & Software Lead in Human-Computer Interaction

Sept. 2024–Present

Louisiana State University, Rutgers University, & University of Zurich

- **NIMH-funded research** with a team of international scholars on a VR-based ADHD intervention study across multiple large participant cohorts. Added and completed novel research objectives, resulting in **2 co-authored peer-reviewed publications**. The study is expanding, and commercialization is advancing via the federal SBIR program.
- Built data pipeline to parse, analyze, and report **180 hours of raw interaction telemetry for the published study's core metrics**, including its strongest predictor of attention; anonymized the dataset for public release.
- **Actively refactoring, security auditing, and expanding a six-app codebase:** Azure-hosted cloud services (a domain-driven FastAPI/PostgreSQL backend, student/manager React web portals, and an OCR microservice) alongside standalone applications (a local telemetry server and a real-time stoplight feedback app).

Software Developer & Systems Administrator (Part Time)

May 2026–Sept. 2026

Henrico County Public Schools

- Sole developer and administrator of the production Flask/PostgreSQL application running CIT's internship program for **48 students, 31 mentors, and 24 partner organizations**.
- Detected, contained, and resolved a security compromise of the production server in a single overnight response; built the first backup infrastructure and a self-service account provisioning service with cross-system LDAP integration.
- Audited, migrated, and decommissioned **25+ internal web applications** from a legacy server onto new production infrastructure, handling networking, nginx configuration, and TLS cutover.

Projects

Project Lead | Woodland Cemetery Restoration Foundation

Jan. 2025–May 2025

Led development team to refactor the organization website, improve mobile functionality, and **map 715+ previously uncharted graves** at a long-neglected historic Richmond, VA cemetery. Displayed graves on an interactive interface showing burial records, 3D grave scans, and Find-A-Grave records. woodlandrestorationfoundation.org/map

Team Captain | "Fintelligent" — Congressional App Challenge Winner

Aug. 2024–Dec. 2024

Led team to win the competitive VA-01 district in a national competition with 3,881 submissions. Our entry was a mobile and desktop app applying image processing and data science to detect pollution-induced disease in freshwater fish; showcased at the U.S. Capitol and met with Congressman Rob Wittman (April 2025). congressionalappchallenge.us/24-VA01/

Mobile Developer & Project Manager | HCPS Transportation

Jan. 2024–May 2024

Co-developed a cross-platform mobile app integrated with the Edulog transportation logistics platform, letting HCPS students verify school-bus boarding via smartphone.

Publications

Chan, E. S. M., da Costa, M., Fritz, T., Frontale, S., Jamalinabijan, F., Cuber, I., **Kurgan, A.**, Shepherd, D. C., & Langberg, J. M. Enhancing Academic Engagement in College Students with ADHD: A Randomized Controlled Trial of a Virtual Reality Study Aid. *Technology, Mind, and Behavior*. **Accepted August 7, 2026.**

Metrics: Q1 Journal; CiteScore: 8; Impact Factor: 3.4

CRedit: Software (Lead), Data Curation (Supporting), Formal Analysis (Supporting)

da Costa, M., Chan, E. S. M., Langberg, J. M., Cuber, I., Jamalinabijan, F., **Kurgan, A.**, Fritz, T., & Shepherd, D. C. Towards Ecological Validity When Assessing ADHD Symptoms: Patterns in Automatically Collected, Real-World PC Activity Data. *International Journal of Human-Computer Studies*, 209, 103724 DOI: [10.1016/j.ijhcs.2025.103724](https://doi.org/10.1016/j.ijhcs.2025.103724)

Metrics: Q1 Journal; CiteScore: 10.1; Impact Factor: 6.1; 16% acceptance

CRedit: Data Curation, Data Analysis, Modeling, Validation, Software Development

Skills

Languages: Proficient in Python, Java, TypeScript/JavaScript, SQL, HTML/CSS, L^AT_EX; C++ (CS 106B, Autumn 2026)
Backend, Web & Tooling: FastAPI, Flask, React.js, REST APIs, PostgreSQL, Git, domain-driven design, microservices
Infrastructure & Security: On-premises Linux administration & hardening, shell/bash, systemd, nginx, Docker, LDAP, backups, incident response, networking, & Azure cloud services
Data & CAD: pandas, NumPy; Onshape (CAD), Fusion 360 (CAM), in-shop machining, 3D printing
Multilingualism: English (Native), Polish (Native), Spanish (Elementary–Intermediate)

Awards & Recognitions

Programming & Cybersecurity

- CyberPatriot National Platinum Division Semi-Finalist 2026, 2025
 - 2026: **2nd in Virginia**, top 3.8% nationally (3,346 teams). Team lead.
 - 2025: 7th in Virginia, **top 1.4% nationally** (5,183 teams).
- Winner, Congressional App Challenge2025
- Finalist, Radford University RUSecure CTF Cybersecurity Competition 2025, 2024

Music

- Top 10 Worldwide, International Leonard Falcone Competition2025
 - One of three high school semi-finalists among university music majors in elite music programs
- Virginia All-State Top Jazz Band, Trombone.....2023–2025
 - Ranked **#1 tenor trombone in Virginia (2024)** & 2nd chair 2023 & 2025, out of 3 statewide top jazz band seats.
- Virginia All-State Band, Euphonium.....2023–2026
 - Concert Band, **2nd chair (2024, 2025)**, 3rd chair (2026); Symphonic Band, 4th chair (2023).
- First Prize, International Classical Music Stars Competition2024

Activities & Leadership

- Robotics | Member of FRC #1086 Blue Cheese** 2022–2024
- Developed 3D robot models using (self-taught) professional CAD software (Onshape). [Example models](#)
 - Fabricated competition robot parts using CAM software (Fusion 360) and in-shop machinery.
 - Led CAD and machining workshops for first-year teammates & served as STEM outreach volunteer at 10+ schools, showcasing robotics, pneumatics, and circuits to elementary students.
- Writing Center Consultant | Southeastern Writing Association Member** 2025–2026
- Selected by the English Department for academic excellence; peer-tutored essays and creative writing weekly, applying methods from the *Bedford Guide for Writing Tutors*.
- Polish School of the Embassy of the Republic of Poland, Washington, D.C.** 2014–2022
- Graduated primary and secondary state-accredited curriculum with distinction — Polish Ambassador’s Diploma for Academic Excellence, 2018–2022.
- Jazz Trombone | Primary Teacher: Toby Whitaker** 2019–Present
- Principal trombone, Greater Richmond Youth Jazz Band (2022–2026).
- Classical Euphonium | Primary Teacher: Prof. Ross A. Walter** 2015–Present
- Frequent solo performer & low brass section leader. [View past performances](#)
 - Lead Euphonium, Central Virginia Wind Symphony (2024) – streamed live on CBS Prime & VPM Music.